

# PFMEA - PFD - Control Plan Linkage Guide

Three-Way Linkage | AIAG-VDA 2019 | Quality Head Agents

## The Three-Way Linkage Requirement

IATF 16949 requires alignment between: Process Flow Diagram (PFD), PFMEA, and Control Plan. Every process step in the PFD must appear in the PFMEA, and every control in the PFMEA must be reflected in the Control Plan.

## Process Flow Diagram (PFD)

The PFD is the foundation. It maps every step in the manufacturing process from raw material receipt to finished goods shipment. Each step has an ID number that links to the PFMEA and Control Plan.

## PFMEA Column by Column

Step 1 (Structure): Process Step Name and Number from PFD. Step 2 (Function): Process requirement. Step 3 (Failure Analysis): Failure modes, effects, causes. Step 4 (Risk): S, O, D ratings, AP. Step 5 (Optimization): Actions for H AP items with verification.

## Control Plan Linkage

For each PFMEA process step: Map detection controls to Control Plan checks. Map prevention controls to Work Instructions. Map special characteristics to Control Plan reaction plans. Document responsible person and frequency.

## Practical Example

Process Step: Heat Treatment (HT-03). PFMEA: Failure Mode = Under-hardness | S=7, O=3, D=2 | AP=M | Prevention: Furnace cert monthly | Detection: 100% hardness test post-HT. Control Plan: 100% Rockwell hardness test, spec 58-62 HRC, reaction = reject and re-treat, notify QA.